

Arush Singh

ZERO-KNOWLEDGE ENGINEER
SMART CONTRACT DEVELOPER

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PROFILE

Blockchain developer specialising in **Solidity Smart Contract** development and **Zero-Knowledge Cryptography**, with a focus on **DeFi Protocols**, on-chain automation, and rigorous testing. Built and deployed production contracts integrating **Chainlink VRF & Automation**, and authored ZK circuits in **Noir** and **Circom** — spanning Merkle Sum Tree proof-of-solvency, undercollateralised credit verification, and Groth16 commit-nullifier voting — with on-chain proof verification via **Barretenberg UltraHonk** and **Groth16** verifiers. Grounded in smart contract security principles — reentrancy, access control, invariant & fuzz testing — and actively advancing toward a **ZK Engineer** specialisation.

SKILLS

SMART CONTRACTS

[Solidity](#) [Foundry](#) [DeFi](#) [Chainlink VRF / Automation / Oracle](#) [OpenZeppelin](#)

SECURITY & TESTING

[Unit Testing](#) [Fuzz Testing](#) [Invariant Testing](#) [Ethernaut](#) [CodeHawks](#)

ZK / CRYPTOGRAPHY

[Noir](#) [Circom](#) [Barretenberg \(bb\)](#) [UltraHonk Verifier](#) [Groth16 / SNARKs](#) [STARKs](#)
[Poseidon](#) [Merkle Sum Trees](#)

TOOLING & LANGUAGES

[Rust](#) [TypeScript](#) [Node.js](#) [React](#) [Linux](#) [Git](#) [Etherscan](#)

PROJECTS

Zero-Knowledge Proof of Solvency

[GitHub](#)

Noir · Merkle Sum Trees · Poseidon · Barretenberg · Solidity · Foundry · TypeScript

- Proof-of-solvency protocol: a custodian proves liabilities coverage and honest per-user inclusion without exposing account data, using a Noir circuit over a **Merkle Sum Tree** where every node commits to a **Poseidon** hash and a running balance sum.
- Range-constrains every balance to $[0, 2^{64})$, closing the **negative-balance attack** that masks a reserve deficit; the TypeScript witness builder is cross-checked against the circuit's Poseidon implementation.
- On-chain UltraHonk verification via `proveAndRecord()` in Solidity/Foundry with **100% line, function, and branch coverage**. Design inspired by Summa (PSE).

ZK-Powered Undercollateralized Lending Prot.

[GitHub](#)

Noir · Barretenberg · Solidity · Foundry · OpenZeppelin · Next.js

- Undercollateralised lending (50% collateral vs the usual 150%): a Noir circuit proves credit score and income exceed thresholds without revealing them, and `LendingPool.borrow()` verifies the proof on-chain via a Barretenberg verifier before funds move.
- 100% coverage** across 23 tests: unit, fuzz, and invariant tests (128,000 actions per invariant) enforcing solvency, accounting integrity, and collateral floors.
- Deployed to Sepolia with a Next.js dApp that generates proofs in-browser via Barretenberg WASM, so private data never leaves the device.

HACKATHONS

ZKVote — Commit-Nullifier Voting DApp

[GitHub](#)

Circom · Groth16 · snarkjs · Poseidon · Solidity · Foundry · React · Node.js

- Voting DApp (final-year project), live on Sepolia at 04arush.github.io/ZKVote: a **Circom** circuit binds a Poseidon commitment to the vote and derives a per-proposal nullifier; **Groth16** proofs are generated in-browser via snarkjs and verified on-chain.
- ZKVoting.sol checks deadline, commitment, and nullifier state before verification; an append-only nullifier registry blocks double-voting. Foundry-tested; React/ethers.js frontend with a Node/MySQL metadata API.

Decentralized Provably Fair Raffle

[GitHub](#)

Solidity · Chainlink VRF v2.5 · Chainlink Automation · Foundry

- Autonomous on-chain raffle with no admin intervention: **Chainlink VRF v2.5** selects the winner and **Chainlink Automation** triggers upkeep.
- Chain-aware deploy script (mocks on Anvil, live Chainlink on Sepolia); 12 unit tests plus fuzz testing of state transitions and access control.

ETHOnline 2026

ethglobal.com/events/ethonline2026

Built **StreamProof**, a private royalty-tier verification protocol: a Noir circuit (Poseidon2 Merkle inclusion + threshold proof) verified on-chain via UltraHonk on Arc Testnet. — Participation Award.

Frostbyte Hackathon — Finalist

frostbyte-hackathon.devpost.com

Qualified with the **Employee Payroll Manager** (Chainlink Automation), then advanced to the finale with the **ZK lending protocol**. — Participation Award.

EDUCATION

Indira Gandhi National Open University (IGNOU)

Aug 2023 - Ongoing

Bachelor's — Computer Application
New Delhi, Delhi, IN

CERTIFICATIONS

Noir Programming & ZK Circuits	Cyfrin Updraft · Apr 2026
Fundamentals of Zero-Knowledge Proofs	Cyfrin Updraft · Mar 2026
Advanced Foundry	Cyfrin Updraft · Mar 2026
Foundry Fundamentals	Cyfrin Updraft · Mar 2026
Rust Programming Basics	Cyfrin Updraft · Feb 2026
Advanced Web3 Wallet Security	Cyfrin Updraft · Feb 2026
Web3 Wallet Security Basics	Cyfrin Updraft · Feb 2026
Solidity Smart Contract Development	Cyfrin Updraft · Jan 2026
Blockchain Basics	Cyfrin Updraft · Dec 2025